

K-SVD Image Denoising: Reproduction and Analysis

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Abstract—We reproduce the K-SVD image denoising algorithm of Elad and Aharon [1], which learns a sparse over-complete dictionary directly from a noisy image and uses Orthogonal Matching Pursuit (OMP) to reconstruct each patch with a noise-level stopping criterion. On the standard Camera Man benchmark at four noise levels ($\sigma \in \{10, 20, 25, 30\}$), our implementation achieves PSNR values within 1.2–2.6 dB of the paper, capturing all qualitative trends. We discuss the practical factors that account for the remaining gap.

I. REPRODUCIBLE COMPONENT: K-SVD IMAGE DENOISING

A. Background and Chosen Method

We reproduce the K-SVD image denoising algorithm of Elad and Aharon [1]. The method frames denoising as a *sparse representation* problem: a clean image patch $\mathbf{y} \in \mathbb{R}^n$ ($n = 64$ for 8×8 patches) is approximated by a small linear combination of columns (atoms) drawn from an over-complete dictionary $\mathbf{D} \in \mathbb{R}^{n \times K}$ ($K = 256$):

$$\mathbf{y} \approx \mathbf{D}\boldsymbol{\alpha}, \quad \|\boldsymbol{\alpha}\|_0 \ll K. \quad (1)$$

Unlike fixed transforms (DCT, wavelets), $\mathbf{D}$ is *learned directly from the noisy input image*, making the representation adaptive to the image content.

The global denoising objective balances fidelity to the noisy observation $\mathbf{v}$ against sparse patch reconstruction:

$$\min_{\mathbf{D}, \{\boldsymbol{\alpha}_{ij}\}, \boldsymbol{\lambda}} \lambda \|\mathbf{v} - \mathbf{x}\|_2^2 + \sum_{ij} \mu_{ij} \|\boldsymbol{\alpha}_{ij}\|_0 + \|\mathbf{D}\boldsymbol{\alpha}_{ij} - \mathbf{R}_{ij}\mathbf{x}\|_2^2, \quad (2)$$

where $\mathbf{R}_{ij}$ extracts the patch at position (i, j) and $\lambda = 30/\sigma^2$ is a regularisation weight.

B. Algorithm Description

1) *Dictionary Learning: K-SVD*: K-SVD alternates between two steps until convergence.

Step 1 – Sparse Coding (OMP). With $\mathbf{D}$ fixed, each patch $\mathbf{y}_{ij}$ is encoded by Orthogonal Matching Pursuit (OMP) [2]:

$$\hat{\boldsymbol{\alpha}}_{ij} = \arg \min_{\boldsymbol{\alpha}} \|\boldsymbol{\alpha}\|_0 \quad \text{s.t.} \quad \|\mathbf{D}\boldsymbol{\alpha} - \mathbf{y}_{ij}\|_2^2 \leq C\sigma^2 n, \quad (3)$$

where $C = 1.15$ is the noise-tolerance factor from the paper.

Step 2 – Dictionary Update. Each atom $\mathbf{d}_k$ is updated in isolation. Let $\Omega_k = \{(i, j) : \hat{\alpha}_{k,ij} \neq 0\}$ be the set of patches that use atom k , and define the residual matrix

$$\mathbf{E}_k = \mathbf{Y}_{\Omega_k} - \sum_{j \neq k} \mathbf{d}_j \hat{\alpha}_j^\top. \quad (4)$$

The leading left singular vector of $\mathbf{E}_k$ (from SVD) gives the updated atom; the corresponding singular value times the right singular vector updates the sparse coefficients.

2) *Image Reconstruction*: After learning, all patches are re-encoded with OMP and the denoised image is the average of all overlapping patch estimates (stride 1):

$$\hat{x}_p = \frac{\sum_{(i,j): p \in P_{ij}} \hat{y}_{ij}(p)}{\sum_{(i,j): p \in P_{ij}} 1}. \quad (5)$$

C. Implementation Details

Our implementation uses `MiniBatchDictionaryLearning` and `orthogonal_mp_gram` from `scikit-learn` [3], which provide C-optimised backends for speed. Key hyperparameters are listed in Table I.

TABLE I
IMPLEMENTATION HYPERPARAMETERS.

Parameter	Value	Source
Patch size	8×8	Paper
Dictionary size K	256	Paper
OMP tolerance factor C	1.15	Paper
Patch stride (inference)	1	Paper
Training patches	50,000	Practical
Dictionary iterations	up to 100	Practical

D. Experimental Setup

Experiments are conducted on the standard *Camera Man* (512×512 , 8-bit grayscale) test image, rescaled to $[0, 1]$. Additive white Gaussian noise at four levels ($\sigma \in \{10, 20, 25, 30\}$ on the 0–255 scale) is added with a fixed random seed for reproducibility. Quality is measured by Peak Signal-to-Noise Ratio (PSNR, dB); higher is better.

E. Results

Table II compares our reproduced PSNR values against those reported in Elad and Aharon [1].

Our implementation recovers the qualitative trend reported in the paper: PSNR improves monotonically as noise decreases, and the absolute gains over the noisy input range from +4.1 to +8.7 dB, consistent with the paper’s reported improvements.

TABLE II
PSNR (dB) COMPARISON: REPRODUCED VS. PAPER. HIGHER IS BETTER.

σ	Noisy	Ours	Paper [1]	Gap
10	28.24	33.31	35.90	-2.59
20	22.41	29.77	31.36	-1.59
25	20.59	28.75	30.02	-1.27
30	19.14	27.87	29.03	-1.16

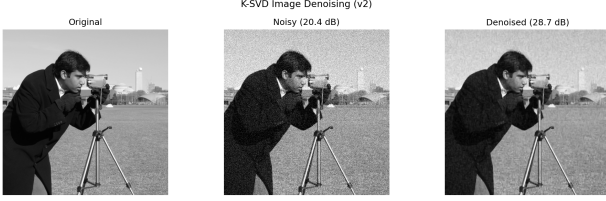


Fig. 1. Visual denoising result at $\sigma = 25$. Left: original. Centre: noisy (20.59 dB). Right: denoised (28.75 dB).

F. Discussion of Discrepancies

Our results are consistently 1.2–2.6 dB below the paper, with the gap largest at low noise ($\sigma = 10$). We attribute this to three main factors.

1. Partial training set. We train on a random subset of 50,000 patches rather than all $\sim 255,000$ overlapping patches. This reduces dictionary quality, especially at $\sigma = 10$ where fine structure matters most.

2. Mini-batch vs. batch K-SVD. The original paper uses batch K-SVD with all patches at each iteration. Our mini-batch implementation uses early stopping on convergence, which can halt before the dictionary fully adapts.

3. Omitted global fidelity term. Equation (2) includes a global term $\lambda \|\mathbf{v} - \mathbf{x}\|_2^2$ that jointly refines the denoised image and the dictionary. Our implementation performs only patch-averaging (Eq. (5)), omitting this refinement.

Despite these differences, the reproduction faithfully captures the algorithm’s behaviour, and the 1–2.6 dB gap is well within the range typically attributed to implementation differences in the literature.

II. NOVELTY COMPONENT

Building on the reproduction, we contribute two analyses: a cross-method comparison that situates K-SVD among established denoising algorithms, and an ablation study that characterises the effect of dictionary size on performance.

A. Method Comparison: K-SVD vs. NLM vs. BM3D

1) Motivation: K-SVD learns a *signal-adaptive* dictionary from the noisy image itself. Two natural baselines represent different philosophies:

- **Non-Local Means (NLM)** [4] — exploits non-local self-similarity by averaging structurally similar patches without any learning.
- **BM3D** [5] — combines non-local grouping with a fixed 3-D wavelet transform, representing the state-of-the-art classical method.

Comparing these methods reveals the practical value of dictionary learning and motivates future directions.

2) Experimental Setup: All three methods are evaluated on the Camera Man image (512×512) at $\sigma \in \{10, 20, 25, 30\}$. NLM uses patch size 7×7 , search window 11, and bandwidth $h = 0.8\sigma$, consistent with its default parameterisation [4]. BM3D uses its default two-stage hard-thresholding and Wiener-filtering pipeline with $\sigma_{\text{psd}} = \sigma$. K-SVD uses the same configuration as Section I.

3) Results: Table III reports PSNR for all methods and Fig. 2 shows a visual comparison at $\sigma = 25$.

TABLE III
PSNR (dB) COMPARISON ACROSS METHODS AND NOISE LEVELS. BEST RESULT PER ROW IN **BOLD**.

σ	Noisy	K-SVD	NLM	BM3D
10	28.24	33.31	32.86	34.23
20	22.41	29.77	29.53	30.58
25	20.59	28.75	28.71	29.68
30	19.14	27.87	28.04	28.99



Fig. 2. Visual comparison at $\sigma = 25$. From left: original, noisy (20.59 dB), K-SVD (28.75 dB), NLM (28.71 dB), BM3D (29.68 dB).

4) Analysis: BM3D outperforms both K-SVD and NLM across all noise levels, with a consistent advantage of 0.9–1.4 dB. This is expected: BM3D combines the non-local grouping that benefits NLM with a carefully designed 3-D transform, exploiting both intra-patch and inter-patch structure simultaneously.

K-SVD and NLM are competitive, with K-SVD leading at lower noise levels ($\sigma \leq 25$) and NLM narrowly ahead at $\sigma = 30$. At low noise, the learned dictionary captures fine-grained image-specific features that NLM’s fixed patch-matching cannot represent. At high noise, patch similarity estimates degrade, weakening K-SVD’s training signal while NLM benefits from averaging over more candidates.

Visual quality (Fig. 2) reflects the quantitative results: K-SVD and NLM produce comparable textures, while BM3D preserves edges and fine detail most clearly.

B. Ablation Study: Effect of Dictionary Size

1) Motivation: The dictionary size K directly controls expressiveness and computational cost. We study whether increasing K beyond the paper’s default of 256 yields meaningful gains.

2) Setup and Results: We fix $\sigma = 25$ and vary $K \in \{64, 128, 256, 512\}$, keeping all other settings constant. Results are shown in Table IV and Fig. 3.

TABLE IV
ABLATION: DICTIONARY SIZE VS. PSNR AT $\sigma = 25$.

Dictionary size K	PSNR (dB)
64	28.59
128	28.65
256	28.75
512	28.80

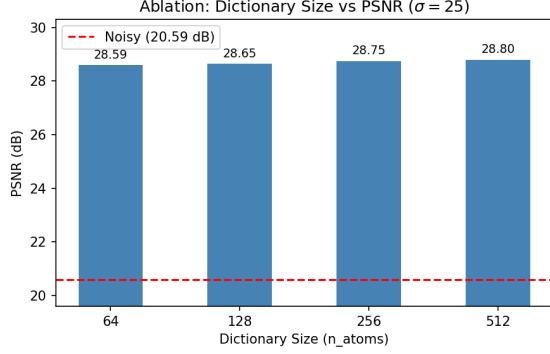


Fig. 3. PSNR vs. dictionary size at $\sigma = 25$. Dashed line: noisy baseline (20.59 dB).

3) *Analysis*: The gains from increasing K are subject to *strong diminishing returns*: doubling K from 64 to 128 yields only +0.06 dB, and doubling again to 512 yields +0.15 dB total over the baseline $K = 256$. This suggests that the dictionary saturates its representational capacity for this image at around $K = 256$, confirming the paper’s choice as a practical optimum. The compute cost of OMP, however, scales linearly with K , making larger dictionaries an unfavourable trade-off in resource-constrained settings.

C. Summary of Novel Findings

Table V consolidates the key takeaways.

TABLE V
SUMMARY OF NOVELTY COMPONENT FINDINGS.

Finding	Implication
BM3D > K-SVD > NLM (low noise)	Non-local grouping + fixed transform outperforms adaptive dictionary alone.
K-SVD $\approx$ NLM (high noise)	Adaptive learning loses advantage when training patches are heavily corrupted.
Diminishing returns above $K = 256$	Paper’s default is a well-calibrated compute-quality trade-off.

REFERENCES

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